

What Graph-Geodesic Embeddings Preserve: Path Fidelity, Scale, and Geometric Recovery

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Abstract

A low-dimensional graph embedding can be assessed through straight-line separations between vertices or through the lengths of paths connecting them. These criteria answer different questions, and neither alone establishes recovery of the geometry that generated the observations. We compare classical scaling, metric stress multidimensional scaling (MDS), and edge-length refinement on sampled paraboloids and saddles. Uniform geodesic bounds show that expanding paraboloids converge, after normalization, to a line metric, while saddles converge to a four-branch tree metric. Under an explicit sample-rank condition, the saddle’s classical-MDS limit spans three Euclidean dimensions despite the one-dimensional intrinsic tree structure. Two controlled studies then separate initialization, graph construction, neighborhood size, and scale policy. Edge refinement improves retained-path agreement under both MDS initializers throughout a bounded-saddle comparison, but the larger radius study exhibits exceptions and substantial differences between path fidelity and coordinate agreement. Profiling the target scale in an unnormalized edge loss permits uniform contraction to decrease the objective; fixed-scale controls separate this effect from changes in relative geometry. The results identify what edge and path diagnostics can establish and why they should be interpreted alongside graph-to-reference error, scale, and initialization sensitivity.

1 Introduction

Graph representations are useful when observations are connected through local similarity and distances along chains of neighboring observations are relevant. For example, an ordination of microbial communities may be interpreted through nearby samples and the routes joining them. A route in such a similarity graph is not automatically a temporal or biological trajectory. Its scientific meaning depends first on the graph construction and then on how the graph is drawn.

Isomap combines neighborhood-graph shortest distances with classical multidimensional scaling (Tenenbaum et al., 2000). Classical scaling constructs a Euclidean representation from a centered squared-dissimilarity matrix; metric stress MDS directly optimizes discrepancies between target distances and Euclidean endpoint distances (R Core Team, 2026; Mair et al., 2022). The two procedures have different objectives. An additional distinction arises when the intended output measurement follows graph paths rather than straight chords.

We study this distinction for a fixed family of input shortest paths and an edge-length refinement referred to here as edge-KK. Exact preservation of edge lengths preserves every path length on the same graph. It does not generally determine the angles between edges or the coordinates of the vertices. Consequently, good path agreement and recovery of a particular surface must be assessed separately. The paper supplies explicit surface limits and paired numerical comparisons that make this distinction measurable.

Classical MDS on general metric measure spaces has been studied through centered distance operators and their spectra (Lim and Mémoli, 2024). Our analytical results concern expanding quadratic surfaces and fixed finite samples, with uniform error bounds that identify the relevant limits. They do not establish arbitrary joint limits in sample size and radius. The numerical studies compare achieved fits, rather than asserting a universal ordering of embedding methods.

Several ingredients of this comparison have established mathematical precedents. Whether edge lengths determine a configuration is a rigidity question (Gortler et al., 2010). Scale sensitivity in graph-drawing stress and the analytic optimization of overall size are addressed by Ahmed et al. (2026). The line and tree quotients also fit the metric Reeb-graph framework, which identifies connected components of function levels and measures variation along the resulting graph (Mémoli et al., 2023). Our contribution is to derive explicit bounds for the two expanding surfaces and connect their MDS behavior to controlled graph, initialization and scale comparisons. We do not propose a new rigidity theory or a new general principle of scale normalization.

2 Distances, objectives, and scale

2.1 A fixed graph and a fixed family of routes

Let $G = (V, E, \ell)$ be a connected undirected graph with positive edge lengths, and let $d_G(i, j)$ denote its shortest-path distance. Choose one shortest path γ_{ij} for each unordered vertex pair, with deterministic tie breaking. For coordinates $Z = (z_1, \dots, z_n)^T \in \mathbb{R}^{n \times p}$, define

$$d_Z(i, j) = \|z_i - z_j\|, \quad L_\Gamma(i, j; Z) = \sum_{(u,v) \in \gamma_{ij}} \|z_u - z_v\|. \quad (1)$$

The triangle inequality gives $d_Z(i, j) \leq L_\Gamma(i, j; Z)$. Equality requires the nonzero displacements along that path to have a common direction. This inequality compares measurements in the same configuration; it does not order their squared errors relative to an external target.

The route family is retained when the coordinates move. Recomputing shortest paths after replacing edge lengths by $\|z_u - z_v\|$ gives another quantity, which is no larger than the retained-route length. Independently selected retained-route lengths need not themselves satisfy all metric axioms.

Write $d_E(Z)$ for the vector of embedded edge lengths and A_Γ for the path-edge incidence matrix. Then $L_\Gamma(Z) = A_\Gamma d_E(Z)$ and $d_G = A_\Gamma \ell$. Thus raw fixed-path stress is

$$S_\Gamma(Z) = \|A_\Gamma(d_E(Z) - \ell)\|^2. \quad (2)$$

The shared edges couple the path residuals. Weighted edge stress generally has a different quadratic form in the edge residual vector.

2.2 MDS and edge refinement

For a symmetric target matrix Δ , raw metric stress is

$$S_\Delta(Z) = \sum_{i < j} (d_Z(i, j) - \Delta_{ij})^2. \quad (3)$$

Equal absolute distance errors make equal contributions to this unweighted loss, regardless of target distance. Equal relative errors make contributions proportional to Δ_{ij}^2 . Long distances therefore matter more for a fixed relative discrepancy, but this observation alone does not predict whether a fitted surface becomes planar or linear. Classical MDS instead retains the largest positive eigenvalues

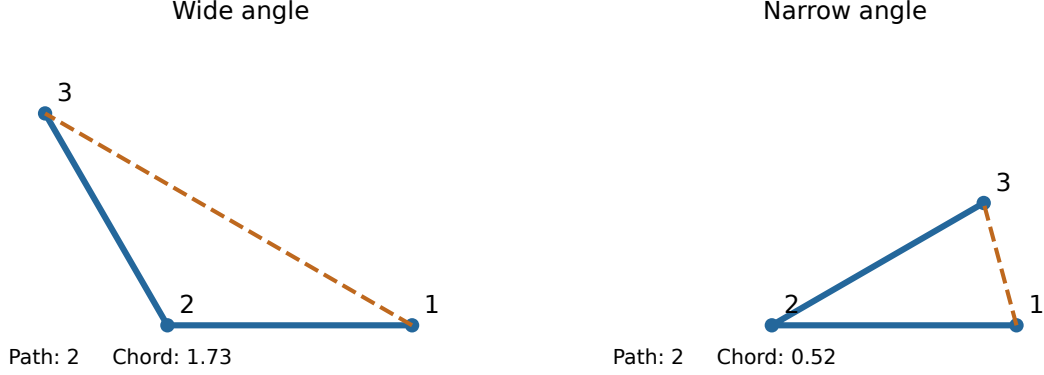


Figure 1: Two drawings of the same three-vertex unit path. Blue segments retain length one, and the route from 1 through 2 to 3 retains length two. The orange endpoint chord changes with the angle. Both drawings have zero edge and fixed-path stress. All spatial units are equal.

and corresponding eigenvectors of $B = -\frac{1}{2}J\Delta^{\circ 2}J$, where $J = I - \mathbf{1}\mathbf{1}^T/n$. Its rank-constrained Gram approximation is commonly called strain (Lim and Mévoli, 2024, Theorem 2). Classical scaling is implemented by `cmdscale` in R (R Core Team, 2026); it does not minimize (3).

Our stress-MDS fits use the ratio-SMACOF implementation in `smacof` (Mair et al., 2022), with the package adapter restoring physical target units. The edge-KK criterion at a given stage is

$$E(Z, a) = \frac{1}{2} \sum_{e \in E} \kappa_e (d_e(Z) - a\ell_e)^2, \quad \kappa_e \geq 0. \quad (4)$$

The experiments vary the stiffness schedule and compare a fixed target scale $a = 1$ with a scale profiled at each configuration. Both are edge objectives; neither directly minimizes (2). For a complete graph with direct-edge targets $\ell_{ij} = \Delta_{ij}$, uniform stiffness and fixed scale, $2E(Z, 1) = S_{\Delta}(Z)$. This identity concerns edge lengths and endpoint chords. A retained route has the same embedded length as its endpoint chord when it is the direct edge; an alternative multiedge route tied in the input need not preserve that equality after fitting.

Proposition 1 (Uniform edge-to-path control). *Suppose $a > 0$ and $|d_e(Z)/(a\ell_e) - 1| \leq \epsilon$ for every edge. For any nonempty path γ in the same graph,*

$$\frac{|\sum_{e \in \gamma} d_e(Z) - a \sum_{e \in \gamma} \ell_e|}{a \sum_{e \in \gamma} \ell_e} \leq \epsilon.$$

Proof. Bound the absolute sum of edge residuals by the sum of their absolute values, then use $|d_e - a\ell_e| \leq \epsilon a\ell_e$ on each edge. $\square$

The premise is a maximum relative error, not an average squared residual. Even zero edge error leaves geometric freedom on flexible graphs. Two unit edges sharing a vertex preserve all fixed-path lengths while their angle changes (Figure 1). At a zero angle their outer endpoints coincide. Exact path fidelity therefore need not imply an injective drawing.

2.3 Profiled scale and contraction

Set $A = \sum_e \kappa_e d_e^2$, $B = \sum_e \kappa_e \ell_e^2$ and $C = \sum_e \kappa_e d_e \ell_e$. If $B > 0$, minimizing (4) over a gives $a(Z) = C/B$ and

$$\mathcal{E}(Z) = \min_a E(Z, a) = \frac{1}{2}(A - C^2/B). \quad (5)$$

Proposition 2 (Homogeneity of the profiled loss). *For $c > 0$, $a(cZ) = ca(Z)$ and $\mathcal{E}(cZ) = c^2\mathcal{E}(Z)$. If $A > 0$, the normalized quantity $Q = 2\mathcal{E}/A = 1 - C^2/(AB)$ is invariant under uniform coordinate scaling.*

Proof. Scaling coordinates by c multiplies A by c^2 and C by c , leaving B unchanged. Substitute these relations into (5). $\square$

A configuration with positive profiled loss can reduce that loss by shrinking, without changing its relative shape. A small loss or gradient can consequently reflect contraction. This issue does not invalidate a separately normalized diagnostic, but it changes the interpretation of optimizer termination. The optimal coordinate-scale calculation below is the weighted least-squares calculation underlying scale-normalized graph stress (Ahmed et al., 2026, Section 3.2); the additional issue here is profiling the target scale while minimizing the unnormalized objective over coordinates.

For comparison, optimize the coordinate scale of a fixed shape against fixed edge targets. If $A > 0$ and $C > 0$, $\min_{c>0} E(cZ, 1) = BQ/2$, attained at $c = C/A$. Thus a properly normalized shape loss and fixed-target fitting with optimal overall size have related shape objectives. The raw profiled loss (5) lacks this normalization. The present experiments use fixed-target controls without introducing a new optimizer for Q .

3 Expanding surfaces and their metric limits

3.1 Paraboloid: a normalized line limit

For $u_i \in \mathbb{R}^2$ in the unit disk, set $q_i = \|u_i\|^2$ and $P_i(r) = (ru_i, r^2 q_i)$ on $z = x^2 + y^2$. Let $g_{ij}(r)$ denote intrinsic geodesic distance on the surface patch above the radius- r disk.

Proposition 3. *For every pair and $r > 0$,*

$$r^2|q_i - q_j| \leq g_{ij}(r) \leq r^2|q_i - q_j| + (1 + \pi)r.$$

Proof. Every surface path has length at least its absolute vertical displacement. For an upper bound, move radially from the outer endpoint to the inner radius, then along the shorter circular arc. Since $\sqrt{1 + 4\rho^2} \leq 1 + 2\rho$, the radial part is at most the height difference plus r , and the arc is at most πr . This path stays in the disk. $\square$

The normalized distances converge uniformly to $|q_i - q_j|$. Angular separation vanishes at this scale; this does not mean its unscaled distance vanishes.

Theorem 4. *Fix a finite sample with at least two distinct q_i and a dimension $p \geq 1$. Every global raw-stress minimizer for $g(r)$, after division of coordinates by r^2 , has a centered Gram matrix converging to $(Jq)(Jq)^T$. The same holds for classical MDS and for approximate raw-stress minimizers with excess stress $o(r^4)$.*

Proof. Writing $Z = r^2 V$ divides the stress by r^4 and the targets by r^2 . The candidate $V_i = (q_i, 0, \dots, 0)$ has normalized stress at most $\binom{n}{2}(1 + \pi)^2/r^2$. Minimizers, including the stated approximate ones, therefore have residuals tending to zero. Their pair distances converge to $|q_i - q_j|$, and double-centering their squared distances gives the Gram limit. For classical MDS, $B(r)/r^4$ converges directly to the same rank-one matrix; its unique positive eigenvalue is separated from zero. $\square$

This is a fixed-sample result. It does not establish that the fitted optimizer outputs attain its approximation requirement. For coupled samples whose $q_i(r)$ converge, the same proof uses the moving line candidate and requires at least two distinct limiting heights.

3.2 Saddle: a tree metric with three-dimensional classical span

Fix (a_i, b_i) in the unit disk. On $z = x^2 - y^2$, write $P_i(r) = (ra_i, rb_i, r^2 h_i)$, where $h_i = a_i^2 - b_i^2$ and $t_i = |h_i|$. Positive heights have separate $+x$ and $-x$ branches; negative heights have separate $+y$ and $-y$ branches. Zero heights map to a common root. Define the four-branch tree distance

$$T_{ij} = \begin{cases} |t_i - t_j|, & i, j \text{ on the same branch,} \\ t_i + t_j, & i, j \text{ on different branches.} \end{cases} \quad (6)$$

Proposition 5. *The intrinsic saddle distances on the disk-restricted patch satisfy $r^2 T_{ij} \leq g_{ij}(r) \leq r^2 T_{ij} + 4r$.*

Proof. Length dominates total vertical variation. Different signs of height require crossing zero. Opposite positive branches require x to cross zero, where height is nonpositive; opposite negative branches similarly require y to cross zero. These facts give the lower bound.

For the upper bound, follow each constant-height hyperbola to its corresponding coordinate semiaxis. On a positive branch, $|dx/dy| \leq 1$ and $|y| \leq r/\sqrt{2}$, so each connector has length at most r ; negative branches exchange x and y . Connect the semiaxis points monotonically on one branch or through the origin for different branches. The inequality used in Proposition 3 bounds this segment by the required vertical variation plus $2r$. All segments remain inside the disk. Zero-height points follow a straight ruling to the origin. $\square$

Thus $g(r)/r^2 \rightarrow T$ uniformly. The limit is intrinsically a one-dimensional tree. Its classical Euclidean approximation need not be one-dimensional.

Theorem 6. *Let F be the $n \times 4$ matrix with row $t_i e_{a(i)}^T$ for a nonroot point, where $a(i)$ is its branch label, and zero row for a root point. If JF has rank four, the centered squared tree-distance matrix has exactly three positive eigenvalues and one negative eigenvalue. Classical three-dimensional MDS of the expanding saddle therefore retains three nonzero limiting singular values after coordinate division by r^2 .*

Proof. Since $t = F\mathbf{1}$, expansion of (6) and double-centering give

$$B_T = -\frac{1}{2} J T^{\circ 2} J = J F (2I_4 - \mathbf{1}\mathbf{1}^T) F^T J. \quad (7)$$

The middle matrix has eigenvalues $2, 2, 2, -2$. The full column rank of JF transfers these nonzero inertia counts to B_T . Proposition 5 implies $B(r)/r^4 \rightarrow B_T$, so the three positive classical directions persist. $\square$

A sufficient rank condition is one nonroot observation per branch and two distinct positive lengths on at least one branch. A root plus one nonroot point per branch also suffices. Missing branches or degenerate samples need separate analysis. The theorem applies to classical scaling; it does not exclude every planar minimizer of raw stress on the tree metric. Non-Euclidean Gram spectra for tree metrics are already illustrated by the three-leaf-star example of [Lim and Mémoli \(2024, Example 2.6\)](#). Here the surface distance bound identifies the tree arising from the expanding saddle, and the rank condition identifies when its classical approximation retains three positive directions.

A simple example makes the spatial dimension concrete. Four branch tips at unit distance from the root have pairwise distances two. If only these tips are observed, their centered matrix is $2J$, and classical MDS represents them exactly as a regular tetrahedron. A planar cross has the right tree routes but not these Euclidean endpoint distances. This tip-only sample has rank three in JF and illustrates why the rank-four condition above is sufficient, not necessary, for a three-dimensional classical span.

Gaussian curvature alone does not resolve this comparison. On these two surfaces, curvature is respectively $4/(1 + 4\rho^2)^2$ and $-4/(1 + 4\rho^2)^2$. Their magnitudes decay identically with radius, yet their normalized distance limits differ because of the connectivity of height levels. [Figure 2](#) shows the corresponding finite-sample behavior.

This explanation can be stated using Reeb graphs. On the unit disk, collapse each connected level component of $q(u) = \|u\|^2$ or $h(a, b) = a^2 - b^2$ to a point. Give a path in the quotient its total variation in function value, as in the metric Reeb construction ([Mémoli et al., 2023, Definitions 2.1 and 3.1](#)). Both height functions are 1-Lipschitz for their normalized surface distances. For q , the quotient is $[0, 1]$ with ordinary distance. For h , positive and negative levels each have two components, joined at the connected zero level; the quotient is the four-branch tree in [\(6\)](#). [Propositions 3 and 5](#) supply explicit uniform distance errors for these particular quotients. The identification uses this established construction, without requiring a general collapse theorem.

4 Controlled experiments

4.1 Bounded-saddle initializer comparison

The first study uses five independent area-uniform samples of $n = 1,000$ points on $z = 0.8(x^2 - y^2)$ over $[-1, 1]^2$. Symmetric-union kNN graphs use ambient chord lengths both for neighbor selection and as edge targets. The selected neighborhood sizes are 71, 70, 67, 72 and 73, based on comparison with numerical surface-geodesic references. This calibration minimizes unscaled graph-to-reference relative RMSE over $k = 3, \dots, 80$, using 119,744 unordered pairs obtained from 128 reference sources and all targets. It uses the known surface and is not a neighborhood-selection rule for unknown biological geometry. Each sample additionally uses $k = 32$, $k = 80$, and the selected $k \pm 5$, giving 25 graphs. These graphs are connected without added bridges.

Classical and raw-stress MDS use strict graph shortest-path matrices and three output dimensions. Stress MDS selects the least achieved stress from six starts: one classical and five Gaussian, with 1,000 iterations per start. Both branches then receive the same five-stage edge-KK schedule, interpolating from density stiffness to uniform stiffness with 200 steps per stage. This historical comparison uses the profiled target scale. The Gaussian draws are paired across k within each sample. Additional iterations are controls and do not replace the primary fits.

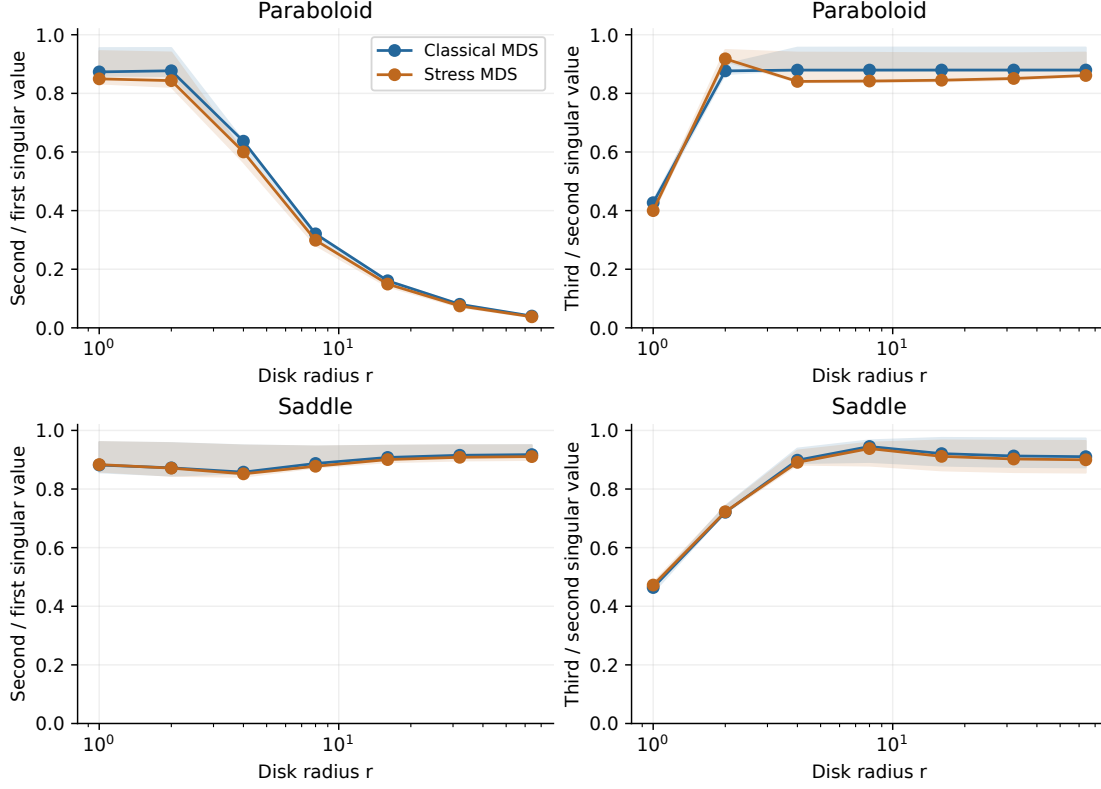


Figure 2: Shape ratios for complete geodesic graphs ($n = 240$, $k = 239$), under base-disk sampling. Curves show medians and bands the observed ranges across three samples. Both ratios are needed: two small transverse directions can remain comparable while the configuration approaches a line. These are classical and finite-budget stress-MDS fits to strict graph distances; numerical smooth-geodesic targets and their strict distances can differ slightly. The classical and global raw-stress statements have the separate assumptions in Theorems 4 and 6.

4.2 Radius, graph, and scale comparison

The second study samples $z = x^2 + y^2$ and $z = x^2 - y^2$ above disks with $r \in \{1, 2, 4, 8, 16, 32, 64\}$. For each surface, both uniform base-disk and uniform surface-area sampling are considered, with three independent $n = 240$ samples and coupled quantiles/angles across radii. No uniform probability distribution on the entire infinite-area surface is assumed.

The neighborhood grid is $k \in \{4, 8, 16, 32, 64, 128, 239\}$. Separate graph regimes use either numerical smooth geodesics or ambient chords for both ranking and edge targets. Graphs are symmetrized by union; minimum-spanning-tree edges repair disconnected cases and are recorded. Classical and stress MDS use each graph's strict distances. Stress MDS uses one classical and two Gaussian starts, each limited to 1,000 iterations. In the geodesic regime, a full-surface-distance classical initializer held fixed across k is an additional control.

Every initializer receives both density continuation and an equal-budget uniform-only refinement, under both profiled and fixed target scales. The primary schedule has the same five mixing values and 1,000 total steps as the first study. Numerical inputs are scaled by the RMS smooth distance within each case and outputs are restored to physical units. The fixed target scale is one in these normalized fitting units.

A nested $n = 480$ sample at $r = 64$ retains the original 240 points of replicate one and uses $k \in \{8, 16, 64, 128, 479\}$. Perturbation, additional-iteration, random-start and original-coordinate controls use 32 selected high-radius settings. Shared original/random controls appear in more than one branch comparison and do not supply independent replications.

4.3 Diagnostics and numerical validation

For positive target vector t and achieved length vector y , set $b = \langle y, t \rangle / \|t\|^2$. Edge and retained-path relative errors are $\|y - bt\| / \|bt\|$, each with its own diagnostic calibration. Chord Stress-1 uses $\|y - bt\| / \|y\|$. These scale conventions are explicit because normalizations called stress are not interchangeable (Mair et al., 2022). Graph-to-surface distance error uses common physical units. Embedded path-to-surface error uses the calibration estimated from graph edges, so it does not fit a new scale against the surface reference.

Coordinate error is a normalized Procrustes residual after translation, orthogonal transformation and one uniform scaling, with vertex correspondence retained. It is not the percentage of misclassified observations. Singular values $s_1 \geq s_2 \geq s_3$ distinguish elongation (s_2/s_1 small) from planarity relative to the second direction (s_3/s_2 small). Neither is a certificate of topology or surface recovery.

The radius study contains 1,216 primary graphs, 16,416 scored primary configurations and 3,648 MDS starts; 74 graphs require connectivity repair. All 88 numerical input matrices pass symmetry, chord lower bounds, constructed surface-path upper bounds and exhaustive vertex-triple checks within 10^{-7} times RMS distance. Independent high-radius numerical validation covers 256 actual pairs in 16 cases, with maximum sampled relative discrepancy 5.16×10^{-8} . This is not a certified all-pairs accuracy bound. The independent reconstruction of diagnostics from 16,928 primary and control coordinate arrays agrees to 2.59×10^{-12} after dimensional scaling. The retained near-tie routes and strict matrices are recorded separately.

5 Results

5.1 Path improvement and geometric agreement

Edge-KK decreases retained-path error in all 25 bounded-saddle settings under both initializers and increases chord Stress-1 in all corresponding comparisons. At the selected graphs, median path error decreases from 3.451% to 0.245% for classical initialization and from 2.264% to 0.164% for stress initialization (Table 1, Figure 3). Stress initialization gives the lower final path error in 23 of 25 graphs, with two reversals retained in the comparisons.

Coordinate agreement does not follow the same pattern. The median coordinate error increases under classical initialization, while decreasing slightly under stress initialization. Across selected graphs, stress-initialized refinement gives coordinate errors ranging from 4.60% to 27.54% despite small final path errors. All recorded edge-KK stages reach their iteration limits. These are achieved finite-budget outcomes, not established minima.

5.2 Graph fidelity and coordinate recovery separate on the saddle

At $r = 64$, $k = 32$, the fixed-scale stress-initialized geodesic saddle fits have median path-to-surface error 0.36% and coordinate error 80.38%. Ambient-graph fits give 0.90% coordinate error but 37.00% path-to-surface error (Table 2). The first configuration preserves intrinsic path measurements much better; the second resembles the generating coordinates more closely. The two outcomes answer different geometric questions. Figure 4 shows the corresponding configurations for the first sample,

Table 1: Bounded-saddle medians over five selected graphs. Path and edge columns use separately calibrated relative errors; chord is profiled Stress-1; coordinate error uses similarity alignment. Edge-KK uses the historical profiled scale.

Method	Path (%)	Edge (%)	Chord (%)	Coordinate (%)
Classical MDS	3.451	7.133	3.459	29.19
Stress MDS	2.264	5.081	2.803	23.20
Classical MDS + edge-KK	0.245	0.382	4.715	30.67
Stress MDS + edge-KK	0.164	0.227	5.213	22.32

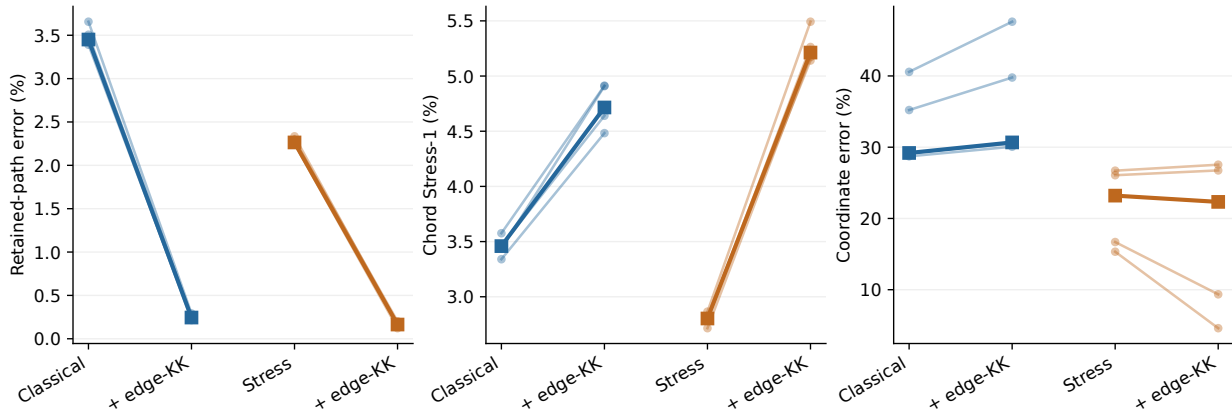


Figure 3: Paired bounded-saddle comparisons at the selected graphs. Thin lines join the same sample before and after edge-KK; squares show medians over five samples. Blue denotes classical initialization and orange stress initialization. Vertical units and ranges differ across the three diagnostics.

together with the smaller-radius comparison. The original surface extends over the entire disk at both radii. Its horizontal extent grows as r and its vertical extent as r^2 , so equal-axis views become narrow at large r . That ambient aspect ratio does not make the intrinsic saddle distance metric a line metric.

The paraboloid has a different pattern: both graph regimes produce strongly elongated configurations at this radius. A small global coordinate error in such a configuration can be dominated by agreement along its largest axis. It should not be interpreted as uniformly accurate angular or transverse geometry. The two shape ratios keep that distinction visible.

5.3 Neighborhood and initialization sensitivity

Graph-to-surface and embedded-path-to-surface errors vary with k (Figure 5). A small graph-to-layout error cannot repair a substantial graph-to-surface error. Even when the full-geodesic classical initializer is held fixed across k , final path-to-surface error changes with the refinement graph in all 84 corresponding $n = 240$ cases.

The nested sample-size check distinguishes fixed k from fixed k/n . For example, under geodesic construction and fixed-scale refinement on the base-disk paraboloid, path-to-surface errors are 7.77% for $(n, k) = (240, 8)$, 9.23% for $(480, 8)$, and 2.34% for $(480, 16)$. This single nested replication does not establish a general sample-size scaling law.

Tiny perturbations around stress-MDS starts change the final s_3/s_2 ratios by at most 0.00031

Table 2: Radius-study medians over three base-disk samples at $r = 64$, $k = 32$, using stress MDS followed by fixed-scale edge-KK. Path/surface error uses edge calibration; coordinate error uses similarity alignment.

Surface	Graph	s_2/s_1	s_3/s_2	Coordinate (%)	Path/surface (%)
Paraboloid	geodesic	0.038	0.846	1.46	0.34
Paraboloid	ambient	0.028	0.915	0.88	0.38
Saddle	geodesic	0.901	0.896	80.38	0.36
Saddle	ambient	0.019	0.917	0.90	37.00

across the tested controls; the corresponding maximum around classical starts is 0.01888. Independently drawn starts can give much larger changes, up to approximately 0.82 under fixed scale. Local stability around an initializer therefore does not imply insensitivity to its selection. Of the 3,648 primary radius-study MDS starts, 25 reach their iteration caps; small stress changes are not a global-optimality certificate.

5.4 Contraction and fixed-scale controls

Across all primary refinements, including both sample sizes, all initializers and both stiffness schedules, the profiled-scale runs attain edge calibration factors as small as 4.07×10^{-10} . The fixed-scale controls range from 0.99339 to 1.02624. Contraction occurs at intermediate radii as well as large ones, so it is not an asymptotic property of the generating surface (Figure 6).

Across the 1,176 primary $n = 240$ graphs, profiled-scale refinement improves path error in 1,016 stress-initialized comparisons, worsens it in 76, and leaves 84 ties. Fixed-scale refinement improves 1,075, worsens 17, and leaves 84 ties. Ties mean absolute changes no larger than 10^{-8} in relative-error units. These counts describe the declared graph grid, not independent trials or probabilities for future data. Neither policy improves path error in every case. Fixed scale removes the uniform-contraction mechanism identified by Proposition 2, while leaving ordinary nonconvexity and geometric ambiguity.

6 Discussion

Graph-geodesic embedding involves at least two approximation problems: the graph approximates a reference geometry, and the drawing approximates selected measurements on that graph. The experiments show why these stages should be reported separately. When the reference surface is unknown, good graph-to-layout agreement establishes only the second comparison. It cannot verify that the chosen graph represents the scientific relationships of interest.

The line and tree limits explain why apparent dimensional collapse cannot be inferred from curvature or a quadratic-fit coefficient alone. A surface’s intrinsic distance limit and the Euclidean span of an MDS approximation are different mathematical objects. For the saddle, a one-dimensional limiting tree produces three positive classical directions under the rank condition. The raw-stress fits are consistent with broad three-dimensional span in the tested settings, but a universal raw-stress theorem has not been established.

Path-length preservation also leaves freedom that is intrinsic to the stated criterion. Adding regularization or a preferred initialization selects among configurations, and such choices should be assessed as additional assumptions. For biological data, the visual placement of a branch or gap should therefore be supported by stability to plausible graph and embedding choices and by evidence

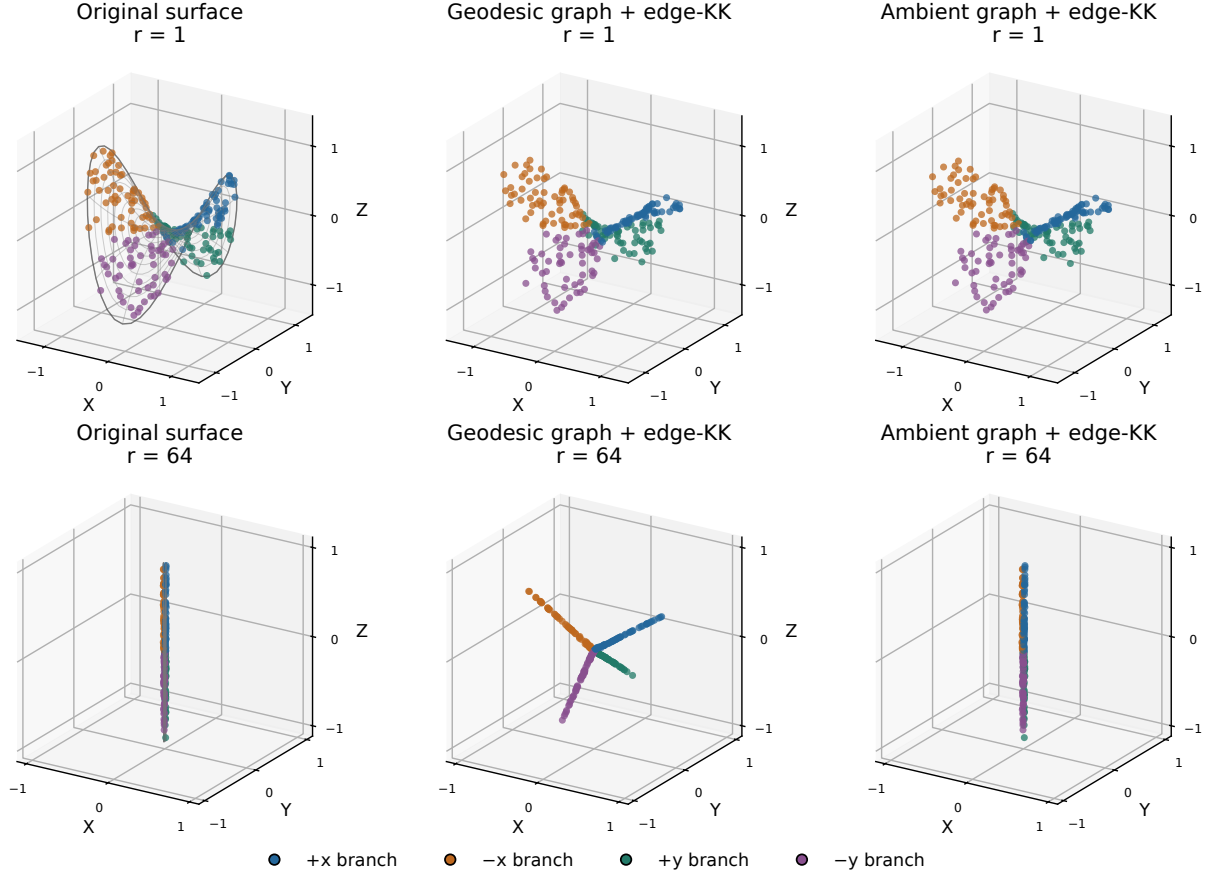


Figure 4: Saddle configurations for the first base-disk sample, $n = 240$, $k = 32$. The fitted columns use stress MDS followed by fixed-scale density continuation. Colors identify the four original height-level branches. The original-surface wireframe includes the full disk boundary. All coordinates are centered and divided uniformly by r^2 ; fitted coordinates are also divided by their edge calibration factor and orthogonally aligned to the original observations. Each row uses common limits, equal spatial axes and an orthographic view. At $r = 64$ the original disk’s horizontal radius in these units is $1/64$, which explains its narrow appearance. No coordinate direction is enlarged independently.

relevant to the biological interpretation. The synthetic experiments do not establish biological trajectories or validate features in a particular microbiome dataset.

The scope is deliberately limited to two quadratic surface families and the stated samples and optimization budgets. Radius varies with fixed sample size in the primary experiment; the samples are not increasingly dense surface meshes. Surface-area and base-disk sampling are distinct. The bounded-saddle study has different geometry, sample size and scale policy from the radius study, so their numerical scores should not be pooled. A larger benchmark or a new regularized optimizer could address further questions, but is not needed to expose the demonstrated differences between path fidelity, scale, and coordinate recovery.

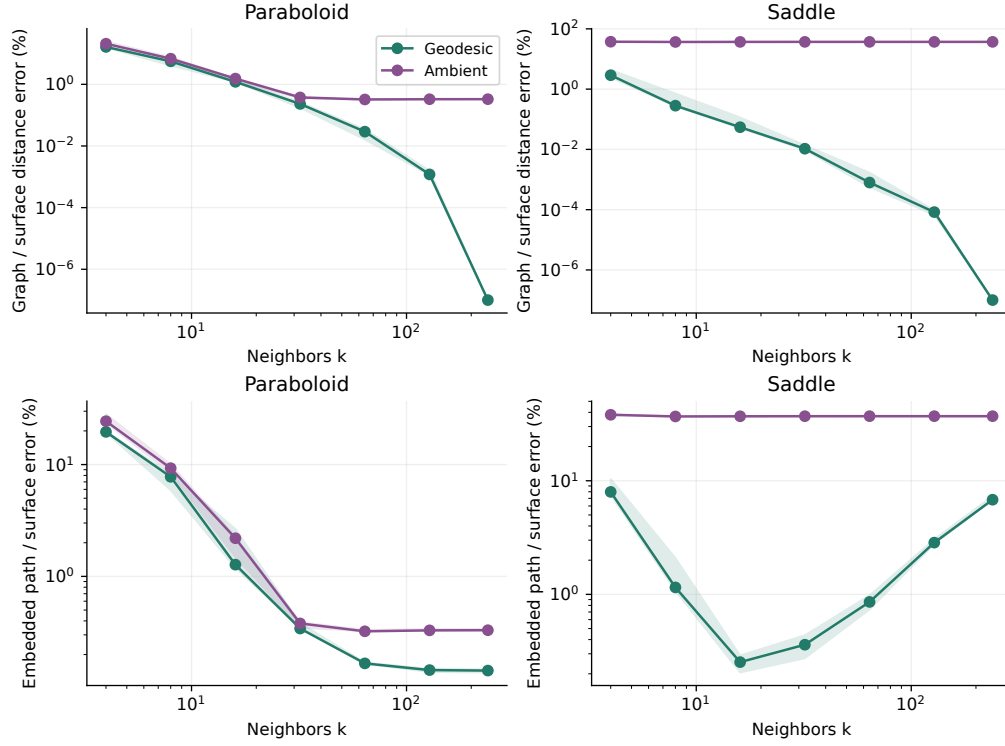


Figure 5: Neighborhood sensitivity at $r = 64$ under base-disk sampling. Curves are medians and bands observed ranges over three samples. The upper row compares strict graph distances with numerical smooth geodesics. The lower row compares edge-calibrated retained paths after fixed-scale, stress-initialized edge-KK with the same reference. Panels use separate logarithmic ranges; values below $10^{-7}\%$ are drawn at that display floor.

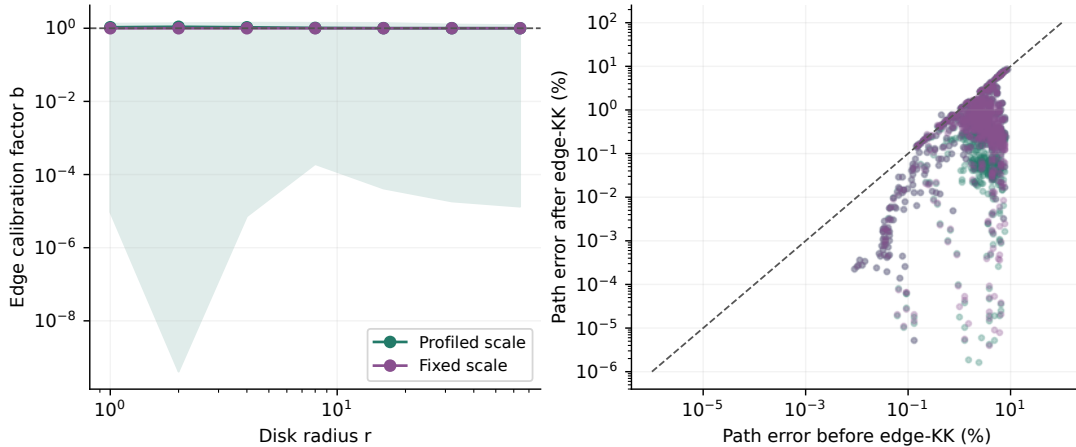


Figure 6: Scale and normalized path fidelity are separate diagnostics. Left: median and full observed range of the edge calibration factor for stress-initialized density continuation across all primary $n = 240$ settings at each radius. Right: paired path errors before and after stress-initialized edge-KK; points below the diagonal improve. Pairs with either error at most 10^{-8} are omitted from this logarithmic scatter only; all comparisons enter the reported counts.

A Computational details

A.1 Sampling and geodesic references

For both radius-study surfaces the area element in polar base coordinates is $\rho\sqrt{1+4\rho^2}d\rho d\theta$. Thus surface-area sampling uses the radial cumulative distribution

$$F_r(\rho) = \frac{(1+4\rho^2)^{3/2} - 1}{(1+4r^2)^{3/2} - 1}, \quad 0 \leq \rho \leq r,$$

with independent uniform angle; base-disk sampling uses $F_r(\rho) = \rho^2/r^2$. The same uniform draws are transformed at each radius. The primary NumPy generator seeds are $20260904 + j$ for replicate j . The nested additional 240 observations use seed 20261905. Corresponding surface/measure settings share these draws, so settings are paired rather than independent samples.

Paraboloid distances use the radial first integral of the geodesic equations, with separate turning and monotone radial paths and 62 bisection steps for the angular boundary condition. Saddle distances use shooting for the Cartesian geodesic equations in base coordinates scaled by r , with continuation from smaller radii. The primary endpoint tolerance is 2×10^{-9} in these base units; the integration relative tolerance is 2×10^{-10} . High-radius validation uses tighter shooting and independent collocation/integration on the saddle, and refined radial solves with adaptive quadrature on the paraboloid. The reported maximum concerns the selected validation pairs, not an error guarantee for all pairs. The bounded-saddle calibration uses a separate triangulated-surface reference, with mesh-refinement and smooth-geodesic boundary-value checks recorded in its original protocol.

A.2 Why complete-surface geodesics stay inside the disk

The smooth solvers do not impose a reflecting or obstacle boundary at the disk edge. This agrees with the stated surface-patch distance. To see this, write $f(x, y) = x^2 + \eta y^2$, $\eta \in \{-1, 1\}$, and let $u = (x, y)$ be the base projection of an affinely parametrized surface geodesic, with $v = u'$ and $\rho^2 = x^2 + y^2$. The graph geodesic equation gives

$$u'' = -\frac{4(v_x^2 + \eta v_y^2)}{1 + 4\rho^2}(x, \eta y),$$

so

$$(\rho^2)'' = 2\|v\|^2 - \frac{8(x^2 + \eta y^2)(v_x^2 + \eta v_y^2)}{1 + 4\rho^2} \geq \frac{2\|v\|^2}{1 + 4\rho^2} \geq 0.$$

Thus the squared radius is convex along a geodesic segment and never exceeds the larger endpoint squared radius. A minimizing geodesic on the complete surface therefore stays in the disk patch whenever its endpoints do.

Both full graphs are complete: their induced metric dominates the Euclidean base metric, and on a bounded base region the two metrics are locally uniformly comparable. The saddle is also simply connected and has negative Gaussian curvature. The Hadamard–Cartan theorem therefore makes its geodesic between two points unique and minimizing (Gorodski, 2022, Theorem 6.5.2, Corollary 6.5.3, and the proof of Lemma 6.5.4, pp. 124–125). This justifies the saddle shooting target; it does not make numerical shooting error zero. The positive curvature of the paraboloid does not permit the same uniqueness argument.

A.3 Stiffness and optimization

At each graph, let $\hat{f}(\ell_e)$ be a Gaussian kernel density estimate of positive input edge lengths, computed using R’s `density` with `nrd0` bandwidth and 512 grid points from the minimum to maximum length. Linear interpolation evaluates the density at each edge length. Write $q_e = \hat{f}(\ell_e)/(|E|^{-1} \sum_f \hat{f}(\ell_f))$. The stiffness at continuation value α is

$$\kappa_e(\alpha) = (1 - \alpha)q_e + \alpha, \quad \alpha \in \{0, 1/4, 1/2, 3/4, 1\}.$$

The implementation normalizes stiffnesses to mean one; no additional signal transformation or positive clipping floor is applied. Equal lengths fall back to uniform stiffness. Uniform-only controls use $\alpha = 1$ for the full budget.

Edge-KK uses gradient descent with Armijo backtracking: initial step one, step multiplier $1/2$, Armijo coefficient 10^{-4} , minimum step 10^{-8} , and Euclidean gradient-norm stopping threshold 10^{-8} . Accepted coordinates are recentered. The edge-length smoothing parameter is zero in both studies. These absolute stopping rules act in each study’s fitting units, which is relevant to the contraction analysis. Stress MDS uses SMACOF tolerance 10^{-8} in both primary studies; a converged start is not necessarily a global minimizer. Radius-study MDS random starts use seed $8300000 + 1000j$, and their draw is reused across graph settings within replicate j .

Neighbor ranks in the radius study order by distance and then vertex index; an edge is present if either endpoint selects the other. Graph edges are canonically ordered before path preparation. The single-route cache records one route per pair. Floating-point near ties in path preparation can yield route lengths slightly different from the strict shortest-distance matrix; the experiments therefore score retained paths against their recorded input route lengths and chords against the strict matrix. Source manifests and the original validators retain this distinction. The mathematical statements for exact shortest paths do not require these discrepancies to be treated as zero.

Data and code availability

The manuscript’s figures and tables are generated from versioned exports of the two grip experiments. A source manifest records the exact commit and SHA-256 hashes of source and exported files. Portable verification and rendering scripts accompany the manuscript. Full fitting protocols, input matrices, coordinates and validators remain in the experiment directories specified by that manifest in the [grip repository](#). The bounded study used grip 0.2.0.9000 and the radius study 0.2.0.9001; both used smacof 2.1.7. The manifest retains their distinct provenance.

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